

# SIMATS ENGINEERING ASSESSMENT TOOL 1

**COURSE NAME:** Computer Networks

**COURSE CODE:** CSA0716

## **COURSE OUTCOME COVERED**

**CO1:** *Apply the functions of OSI layer in enabling reliable data transmission across networks.  
(BL2, BL3)*

*Assessment Tool Weightage: 40%*

## Annexure A

### **SHORT ANSWER TEST**

#### ***Code of Conduct:***

*I Naga Jaswanth B (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** Naga Jaswanth B

# COMPUTER NETWORKS (CSA0716)

## Assessment Tool 1 – Short Answer Test Solutions

### QUESTION 1

#### Given:

- File size = 150 MB = 150,000,000 bytes
- Segment size = 1500 bytes
- Data Link Layer overhead = 40 bytes per frame
- Bandwidth = 50 Mbps

#### Calculation

##### Step 1: Calculate the number of segments

Number of segments = File size  $\div$  Segment size

$$= 150,000,000 \div 1500$$

$$= \mathbf{100,000 \text{ segments}}$$

Since one segment is placed into one frame,

$$\text{Number of frames} = \mathbf{100,000 \text{ frames}}$$

##### Step 2: Calculate the total Data Link Layer overhead

Total overhead = Number of frames  $\times$  Overhead per frame

$$= 100,000 \times 40$$

$$= 4,000,000 \text{ bytes}$$

$$= 4 \text{ MB}$$

### Step 3: Calculate the total data transmitted

Total data transmitted

$$= \text{Original file size} + \text{Overhead}$$

$$= 150 \text{ MB} + 4 \text{ MB}$$

$$= 154 \text{ MB}$$

### Step 4: Calculate the transmission time

Convert MB into Megabits

$$154 \times 8$$

$$= 1232 \text{ Megabits}$$

Transmission time

$$= \text{Total bits} \div \text{Bandwidth}$$

$$= 1232 \div 50$$

$$= 24.64 \text{ seconds}$$

**Answer**

- Number of segments = **100,000**

- Number of frames = **100,000**
- Total overhead = **4 MB**
- Transmission time = **24.64 seconds**

### **Explanation**

The **Transport Layer** divides the file into smaller segments and ensures reliable delivery using sequence numbers, acknowledgements, retransmissions, and flow control. The **Data Link Layer** encapsulates each segment into frames by adding headers and trailers, performs error detection using CRC, and ensures reliable communication between directly connected devices. Together, these layers provide accurate, efficient, and reliable data transmission.

### **QUESTION 2**

#### **Given:**

- Sliding window size = 6 frames
- Total frames = 48
- Frame size = 1024 bytes

#### **Calculation**

##### **Step 1: Calculate the number of transmission windows**

Number of windows

= Total frames ÷ Window size

= 48 ÷ 6

= **8 windows**

### **Step 2: Calculate retransmissions**

One frame is lost in every window.

Total retransmissions

$$= 8 \times 1$$

$$= \mathbf{8 \text{ frames}}$$

### **Answer**

- Number of transmission windows = **8**
- Total retransmissions = **8 frames**

### **Explanation**

Flow control regulates the rate of data transmission between the sender and receiver. It prevents receiver buffer overflow, minimizes packet loss, avoids unnecessary retransmissions, and improves communication efficiency. The sliding window protocol allows multiple frames to be transmitted before receiving acknowledgements, thereby increasing network throughput.

### **QUESTION 3**

#### **Given:**

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

## Calculation

Maximum payload in one fragment

$$= \text{MTU} - \text{Header}$$

$$= 1500 - 20$$

$$= \mathbf{1480 \text{ bytes}}$$

Payload to be fragmented

$$= 12,000 - 20$$

$$= \mathbf{11,980 \text{ bytes}}$$

Number of fragments

$$= 11,980 \div 1480$$

$$= 8.09$$

Therefore,

$$= \mathbf{9 \text{ fragments}}$$

Header overhead

$$= \text{Number of fragments} \times \text{Header size}$$

$$= 9 \times 20$$

$$= \mathbf{180 \text{ bytes}}$$

### Answer

- Number of fragments = **9**
- Total header overhead = **180 bytes**

### Explanation

The **Network Layer** performs fragmentation whenever a packet exceeds the MTU of the network. Each fragment contains its own IP header with identification, fragment offset, and control information. At the destination, these fragments are reassembled correctly to reconstruct the original packet before it is delivered to the upper layers.

### QUESTION 4

#### Given:

- Sliding window size = 6 frames
- Total frames = 48

#### Calculation

##### Step 1: Number of windows

$$48 \div 6$$

$$= \mathbf{8 \text{ windows}}$$

##### Step 2: Retransmissions

One frame is lost in every window.

Total retransmissions

$$= 8 \times 1$$

**= 8 frames**

### **Answer**

- Number of windows = **8**
- Total retransmissions = **8 frames**

### **Explanation**

Flow control synchronizes the transmission speed between the sender and receiver. It prevents congestion, reduces frame loss, improves bandwidth utilization, and ensures efficient communication using the sliding window protocol.

### **QUESTION 5**

#### **Given:**

- Video size = 600 MB
- Synchronization checkpoint every 60 MB
- Failure occurs after 390 MB

#### **Calculation**

##### **Step 1: Calculate completed checkpoints**

$$390 \div 60$$

$$= 6.5$$

Completed checkpoints

$$= 6$$

**Step 2: Last successful checkpoint**

$$6 \times 60$$

$$= 360 \text{ MB}$$

**Step 3: Data to be retransmitted**

$$390 - 360$$

$$= 30 \text{ MB}$$

**Answer**

- Number of checkpoints created = **6**
- Data retransmitted after recovery = **30 MB**

**Explanation**

The **Session Layer** establishes, manages, and terminates communication sessions. Synchronization checkpoints divide long transmissions into recovery points. When a failure occurs, transmission resumes from the last checkpoint instead of restarting from the beginning. This reduces retransmission time, saves bandwidth, and improves communication reliability.

**QUESTION 6**

**Given:**

- Original file size = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%** of the compressed file size

**Calculation**

**Step 1: Calculate the compressed file size**

Compressed file size

$$= \text{Original file size} \times (100 - \text{Compression}) \div 100$$

$$= 900 \times 60 \div 100$$

$$= \mathbf{540 \text{ MB}}$$

**Step 2: Calculate the encryption overhead**

Encryption overhead

$$= 540 \times 5 \div 100$$

$$= \mathbf{27 \text{ MB}}$$

**Step 3: Calculate the encrypted file size**

Encrypted file size

$$= \text{Compressed file size} + \text{Encryption overhead}$$

$$= 540 + 27$$

$$= 567 \text{ MB}$$

#### Step 4: Calculate the total percentage reduction

Percentage reduction

$$= ((\text{Original file size} - \text{Encrypted file size}) \div \text{Original file size}) \times 100$$

$$= ((900 - 567) \div 900) \times 100$$

$$= (333 \div 900) \times 100$$

$$= 37\%$$

#### Answer

- Compressed file size = **540 MB**
- Encrypted file size = **567 MB**
- Total percentage reduction = **37%**

#### Explanation

The **Presentation Layer** is responsible for data compression, encryption, and format conversion. Compression reduces the amount of data transmitted, improving bandwidth utilization. Encryption protects data from unauthorized access during transmission. Thus, the Presentation Layer enhances both communication efficiency and security.

#### QUESTION 7

**Given:**

- Data transferred = **3 GB**
- Effective throughput = **120 Mbps**
- Request size = **3 MB**

### **Calculation**

#### **Step 1: Convert GB into MB**

$$3 \text{ GB} = \mathbf{3000 \text{ MB}}$$

#### **Step 2: Calculate the number of requests**

Number of requests

$$= \text{Total data} \div \text{Request size}$$

$$= 3000 \div 3$$

$$= \mathbf{1000 \text{ requests}}$$

#### **Step 3: Calculate the transmission time**

Convert data into Megabits

$$3000 \text{ MB} \times 8$$

$$= \mathbf{24,000 \text{ Megabits}}$$

Transmission time

$$= \text{Total Megabits} \div \text{Throughput}$$

$$= 24,000 \div 120$$

= **200 seconds**

### **Answer**

- Number of requests = **1000**
- Transmission time = **200 seconds**

### **Explanation**

The **Application Layer** provides network services directly to end users. FTP enables reliable file transfer by managing file requests, user authentication, and communication between the client and server. It ensures accurate, secure, and efficient file transfer over the network.

### **QUESTION 8**

#### **Given:**

- Application Layer delay = **4 ms**
- Transport Layer delay = **3 ms**
- Network Layer delay = **5 ms**
- Data Link Layer delay = **2 ms**
- Physical Layer delay = **1 ms**
- Propagation delay = **18 ms**
- Transmission delay = **12 ms**

#### **Calculation**

### **Step 1: Calculate the total processing delay**

Processing delay

$$= 4 + 3 + 5 + 2 + 1$$

$$= \mathbf{15\ ms}$$

### **Step 2: Calculate the total end-to-end delay**

Total delay

$$= \text{Processing delay} + \text{Propagation delay} + \text{Transmission delay}$$

$$= 15 + 18 + 12$$

$$= \mathbf{45\ ms}$$

### **Answer**

- Processing delay = **15 ms**
- Total end-to-end delay = **45 ms**

### **Explanation**

Each OSI layer contributes to successful communication. The Application Layer provides user services, the Transport Layer ensures reliable delivery, the Network Layer performs routing, the Data Link Layer manages framing and error detection, and the Physical Layer transmits bits through the communication medium. Together, these layers determine the total communication delay.

## QUESTION 9

**Given:**

- Useful data = **80 MB = 80,000,000 bytes**
- Frame size = **1600 bytes**
- Protocol overhead = **80 bytes per frame**

### Calculation

**Step 1: Calculate the useful data per frame**

Useful data per frame

= Frame size – Overhead

= 1600 – 80

= **1520 bytes**

**Step 2: Calculate the number of frames**

Number of frames

= Total useful data ÷ Useful data per frame

= 80,000,000 ÷ 1520

= 52,631.58

≈ **52,632 frames**

**Step 3: Calculate the total protocol overhead**

Total overhead

= Number of frames  $\times$  Overhead per frame

=  $52,632 \times 80$

= **4,210,560 bytes**

#### **Step 4: Calculate the transmission efficiency**

Transmission efficiency

= (Useful data per frame  $\div$  Frame size)  $\times$  100

=  $(1520 \div 1600) \times 100$

=  $0.95 \times 100$

= **95%**

#### **Answer**

- Number of frames = **52,632**
- Total protocol overhead = **4,210,560 bytes**
- Transmission efficiency = **95%**

#### **Explanation**

Protocol overhead consists of headers and trailers added by different OSI layers. Although overhead reduces the effective data rate slightly, it is necessary for addressing, routing, error detection, and reliable communication. Efficient protocol design minimizes overhead while maintaining network performance and reliability.

## QUESTION 10

### Given:

- Original medical record size = **240 MB**
- Compression = **30%**
- Segment size = **1400 bytes**
- Data Link Layer overhead = **60 bytes per frame**

### Calculation

#### Step 1: Calculate the compressed file size

Compressed file size

$$= \text{Original size} \times (100 - \text{Compression}) \div 100$$

$$= 240 \times 70 \div 100$$

$$= \mathbf{168 \text{ MB}}$$

#### Step 2: Convert compressed file into bytes

168 MB

$$= \mathbf{168,000,000 \text{ bytes}}$$

#### Step 3: Calculate the number of segments

Number of segments

$$= 168,000,000 \div 1400$$

$$= \mathbf{120,000 \text{ segments}}$$

#### **Step 4: Calculate the total protocol overhead**

Protocol overhead

$$= \text{Number of segments} \times \text{Overhead per frame}$$

$$= 120,000 \times 60$$

$$= \mathbf{7,200,000 \text{ bytes}}$$

$$= \mathbf{7.2 \text{ MB}}$$

#### **Step 5: Calculate the final amount of data transmitted**

Final transmitted data

$$= \text{Compressed file size} + \text{Protocol overhead}$$

$$= 168 + 7.2$$

$$= \mathbf{175.2 \text{ MB}}$$

#### **Answer**

- Compressed file size = **168 MB**
- Number of segments = **120,000**
- Total protocol overhead = **7.2 MB**

- Final data transmitted = **175.2 MB**

### **Explanation**

The **Presentation Layer** compresses the medical record to reduce its size and may encrypt it to ensure confidentiality. The **Transport Layer** divides the compressed data into segments for reliable transmission. The **Data Link Layer** encapsulates each segment into frames by adding headers and trailers for error detection. Finally, the **Physical Layer** transmits the data through the communication medium. Together, these layers provide secure, reliable, and efficient communication in an IoT-enabled healthcare network.